

Planning and Assessment

Series: M2S | **Notebook:** 3 of 8 | **Created:** January 2026 | **Last Updated:** 02/06/2026

The Plan phase is critical—most migration problems stem from incomplete discovery or unclear strategy. This notebook guides you through systematic planning.

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Prerequisites

Before starting this notebook, you should have:

Requirement	Description
Completed M2S-01 & M2S-02	Framework understanding
API Token	With <code>entities.read</code> , <code>settings.read</code> , <code>metrics.read</code> scopes
Admin Access	To Dynatrace Managed environment
Spreadsheet Tool	For inventory documentation

Learning Objectives

By the end of this notebook, you will:

- Complete a thorough discovery of your Managed environment
 - Document all configurations requiring migration
 - Define success criteria for your migration
 - Create a risk assessment matrix
 - Have a preliminary migration plan
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1. Introduction

Why Planning Matters

Planning Gap	Consequence
Missing host inventory	Incomplete migration
Unknown integrations	Broken workflows
Unclear timeline	Resource conflicts
No success criteria	Uncertain completion

2. Discovery: Inventory Your Environment

2.1 Entity Inventory

Important: DQL (Grail) is only available in SaaS environments. For discovering your Managed environment, use the **Entities API v2** or the **Dynatrace UI**.

Use the Entities API to build your entity inventory:

```
# Complete host inventory with details
curl -X GET "https://{your-managed-url}/api/v2/entities?
entitySelector=type(HOST)&fields=properties.osType,properties.oneAgentVer
sion,properties.cloudType" \
-H "Authorization: Api-Token {TOKEN}" &gt; hosts.json

# Service inventory
curl -X GET "https://{your-managed-url}/api/v2/entities?
entitySelector=type(SERVICE)&fields=properties.serviceType" \
-H "Authorization: Api-Token {TOKEN}" &gt; services.json

# Application inventory
curl -X GET "https://{your-managed-url}/api/v2/entities?
entitySelector=type(APPLICATION)" \
-H "Authorization: Api-Token {TOKEN}" &gt; applications.json

# Synthetic monitor inventory
curl -X GET "https://{your-managed-url}/api/v2/entities?
entitySelector=type(SYNTHETIC_TEST)" \
-H "Authorization: Api-Token {TOKEN}" &gt; synthetics.json
```

Or navigate in the Dynatrace UI:

- **Hosts:** Infrastructure → Hosts
- **Services:** Applications & Microservices → Services
- **Applications:** Applications & Microservices → Frontend
- **Synthetics:** Digital Experience → Synthetic

Discovery Checklist		
Entities	Configurations	Integrations
☐ Hosts (count by OS)	☐ Management Zones	☐ ITSM (ServiceNow)
☐ Services (by type)	☐ Auto-tagging Rules	☐ Notifications
☐ Applications	☐ Alerting Profiles	☐ CI/CD Pipelines
☐ Process Groups	☐ Dashboards	☐ Custom Webhooks
☐ Synthetic Monitors	☐ SLOs	☐ SSO/SAML Config
☐ ActiveGates	☐ Request Attributes	☐ API Consumers
☐ OneAgent Versions	☐ Maintenance Windows	☐ User Permissions

3. Strategy: Define Your Approach

3.1 Migration Approach Decision

Based on your discovery, choose your approach:

Your Environment	Recommended Approach
< 500 hosts, simple integrations	Big Bang
500–2000 hosts, moderate complexity	Phased by environment
> 2000 hosts, complex integrations	Phased by region/application
Strict compliance requirements	Phased with extended validation

3.2 Migration Tooling Decision

Choose your primary migration tooling early—**do not mix approaches**:

Tool	Best For	Prerequisites
SaaS Upgrade Assistant	Most migrations—UI-based, automated	Managed v1.294+, same major version alignment
Monaco	Config-as-code environments, tenant consolidation	Monaco CLI installed, environment YAML files
Terraform	IaC-driven organizations	Dynatrace Terraform provider configured
Settings API	Custom scripted migrations	API tokens with read/write scopes

Important: The [SaaS Upgrade Assistant](#) is the recommended tool for most migrations. It handles export, import, dependency management, and progress tracking in a single app. However, if you are consolidating multiple Managed tenants with **identical configuration names**, use Monaco instead—the SaaS Upgrade Assistant may have difficulty resolving name conflicts in this scenario.

3.3 Version Alignment Planning

If using the SaaS Upgrade Assistant, verify version compatibility:

Check	Requirement
Managed cluster version	Must be 1.294 or later for SaaS Upgrade Assistant support
Version alignment	Same major version recommended (e.g., Managed 1.294.x ↔ SaaS 1.294.x)
Impact of mismatch	Mismatched versions cause false-positive configuration failures during import

Tip: Check your Managed cluster version in Cluster Management Console → Overview. Plan a Managed cluster update if needed before starting the SaaS Upgrade Assistant export.

3.4 Licensing Considerations

Important: Dual licensing and contract alignment are critical considerations for migration planning.

Consideration	Description
Dual-run period	Both Managed and SaaS may run simultaneously during migration
License overlap	Coordinate with Dynatrace for temporary overlap licensing
Contract timing	Align SaaS contract start with migration timeline
Host counting	Hosts appear in both environments during dual-run

Work with your Dynatrace account team to:

- Clarify licensing for the migration period
- Align contract terms with migration timeline
- Understand any temporary cost implications
- Plan the decommissioning of Managed licenses

3.5 Historic Data Considerations

 **Critical:** Historic monitoring data cannot be migrated from Managed to SaaS.

Data Type	Migration Status
Metrics history	✗ Cannot migrate
Log history	✗ Cannot migrate
Trace history	✗ Cannot migrate
Problem history	✗ Cannot migrate
User session data	✗ Cannot migrate
Configurations	✓ Can migrate via SaaS Upgrade Assistant or API

Planning implications:

- Start SaaS data collection before decommissioning Managed
- Consider dual-run period for baseline comparison
- Export any critical reports before shutdown
- Maintain Managed access for historical reference (if needed)

3.6 Success Criteria Definition

Define measurable success criteria:

Criterion	Target	Measurement
Host coverage	100% of current hosts reporting	Entity count in SaaS
Service discovery	All services detected	Service count in SaaS
Data continuity	No gaps > 15 minutes	Metrics analysis
Alert functionality	Test alerts received	Notification test
Dashboard accuracy	All tiles showing data	Visual inspection
Integration health	All webhooks functioning	Integration tests
Config migration	All deployable configs imported	SaaS Upgrade Assistant deploy results

3.7 Timeline Planning

Phase	Duration	Key Milestones
Plan	Week 1-2	Discovery complete, strategy approved, tooling selected
Prepare	Week 3	SaaS tenant ready, SaaS Upgrade Assistant installed, network configured
Execute	Week 4	Config exported/imported via SaaS Upgrade Assistant, agents migrated
Validate	Week 5	All success criteria met, deploy results reviewed
Optimize	Week 6+	Tuning and feature adoption

Best Practice: Minimize the dual-run period to reduce complexity and cost, but allow enough time for proper validation.

3.8 Stakeholder Matrix

Stakeholder	Role	Communication Frequency
Executive Sponsor	Approval, escalation	Weekly status
Platform Team	Technical execution	Daily standups
Application Teams	Validation, feedback	Phase notifications
Security Team	Compliance sign-off	Design review, final approval
Dynatrace Account Team	Support, guidance	As needed

4. Design: Create Your Plan

4.1 Migration Order of Operations

Follow this proven sequence for successful migration:

Step	Phase	Activity	Details
1	Plan	Assess	Inventory current environment - entities, configs, integrations
2	Prepare	Provision	SaaS tenant and access (SSO setup, IAM configuration)
3	Prepare	Install	New ActiveGates in parallel with old ActiveGates
4	Execute	Migrate	Configuration and integrations (via API/tooling)
5	Execute	Rebuild	Dashboards, management zones, alerts (non-portable items)
6	Execute	Redirect	OneAgents to SaaS (repoint or reinstall)
7	Execute	Reconnect	Integrations and extensions (validate connectivity)
8	Execute	Migrate	Any remaining configuration and integrations
9	Validate	Validate	Data flow, performance, coverage
10	Cutover	Cutover	Full switch to SaaS, notify stakeholders
11	Run	Decommission	Managed environment shutdown

Note: Steps 4-8 may overlap and iterate. The sequence ensures dependencies are satisfied (e.g., configurations in place before agents redirect).

4.2 Network Design

Component	Current (Managed)	Target (SaaS)
Cluster endpoint	Internal cluster URL	<code>{tenant}.live.dynatrace.com</code>
OneAgent routing	Cluster ActiveGate	Environment ActiveGate or direct
Firewall rules	Internal only	Outbound 443 to SaaS

4.3 ActiveGate Strategy

Scenario	Recommendation
Direct internet access	OneAgent direct to SaaS
Proxy required	Configure proxy on OneAgent
No internet access	Environment ActiveGate for routing
Multi-region	Regional Environment ActiveGates

4.4 Configuration Migration Sequence

Migrate configurations BEFORE redirecting OneAgents:

Order	Configuration Type	Why This Order
1	Deep monitoring settings	Affects how services are detected
2	Custom process monitoring rules	Affects process group detection
3	Container injection rules	Affects Kubernetes monitoring
4	Management zones	Needed for proper organization
5	Auto-tagging rules	Needed for filtering/organizing
6	Service detection rules	Affects service discovery
7	Request attributes	Needed for request data capture

Migrate AFTER OneAgents redirect (entities must exist):

Order	Configuration Type	Why This Order
8	Entity-level settings	Requires entity to exist in SaaS
9	Dashboards	Need data to validate
10	Alerting profiles	Need entities for filtering
11	Problem notifications	Need integrations validated

4.5 Rollback Plan

Scenario	Rollback Action	Recovery Time
OneAgent issues	Reconfigure to Managed endpoint	Minutes
Data quality issues	Pause migration, investigate	Hours
Complete failure	Full rollback to Managed	Hours

5. Risk Assessment

5.1 Risk Matrix

Risk	Likelihood	Impact	Mitigation
Network connectivity issues	Medium	High	Pre-test connectivity, have rollback
Configuration mismatch	Medium	Medium	Use SaaS Upgrade Assistant preview, validate before deploy
Integration failures	Medium	High	Test integrations early, document endpoints
Data gaps during migration	Low	Medium	Parallel operation period
User access issues	Low	Medium	SSO configuration testing
Performance degradation	Low	High	Monitor during migration, capacity planning
Version mismatch (SaaS Upgrade Assistant)	Medium	Medium	Align Managed cluster to same major version as SaaS

5.2 Go/No-Go Checklist

Before proceeding to Execute phase:

Checkpoint	Status
Discovery documented	[]
Strategy approved	[]
Migration tooling selected	[]
SaaS Upgrade Assistant installed (if applicable)	[]
Managed cluster version verified (1.294+)	[]
Network connectivity verified	[]
SaaS tenant provisioned	[]
API tokens created	[]
Rollback plan documented	[]
Stakeholders notified	[]
Maintenance window scheduled	[]

6. Next Steps

Immediate Actions

1. **Complete discovery API calls** - Run all API commands in this notebook
2. **Document configurations** - Create configuration inventory spreadsheet
3. **Select migration tooling** - [SaaS Upgrade Assistant](#) recommended for most scenarios
4. **Verify Managed version** - Ensure cluster is v1.294+ for SaaS Upgrade Assistant compatibility
5. **Define success criteria** - Get stakeholder agreement
6. **Create timeline** - Schedule phases and milestones
7. **Identify risks** - Complete risk assessment

Continue the Series

Next Notebook	Focus
M2S-04: Architecture & Design	Network topology and ActiveGate placement

Planning Resources

- [SaaS Upgrade Assistant Documentation](#)
- [Upgrading from Dynatrace Managed to SaaS](#)
- [Dynatrace API Documentation](#)
- [Settings API Reference](#)
- [ActiveGate Sizing](#)

Summary

In this notebook, you learned:

- How to conduct thorough environment discovery using the Entities API
- API calls for building entity inventory from Managed environments
- How to select the right migration tooling (SaaS Upgrade Assistant recommended)
- Version alignment requirements for the SaaS Upgrade Assistant
- Configuration categories requiring migration
- How to define success criteria and timeline
- Risk assessment methodology

Key Takeaway: Thorough planning prevents migration problems. Invest time in discovery—every unknown configuration is a potential issue during execution. Select the [SaaS Upgrade Assistant](#) as your primary tool early, and verify version compatibility before starting.

Continue to M2S-04: Architecture & Design for detailed technical planning.

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